

# CALCULUS (I)

## EXERCISE SHEET (I)

### LIMITS AND CONTINUITY

① Evaluate the following limits

②  $\lim_{x \rightarrow -4} \frac{x^2 + 5x + 4}{x^2 + 3x - 4} = ?$  b

$$\lim_{x \rightarrow -4} \frac{(x+1)(x+4)}{(x+4)(x-1)} = \lim_{x \rightarrow -4} \frac{(x+1)}{(x-1)} = \frac{-3}{-5} = \frac{3}{5}$$

③  $\lim_{x \rightarrow 4} \frac{x^2 - 4x}{x^2 - 3x - 4} = ?$

$$\lim_{x \rightarrow 4} \frac{x(x-4)}{(x-4)(x+1)} = \frac{4}{5}$$

④  $\lim_{t \rightarrow -3} \frac{t^2 - 9}{2t^2 + 7t + 3} = \lim_{t \rightarrow -3} \frac{(t-3)(t+3)}{(2t+1)(t+3)}$   
 $= \frac{-6}{-5} = \frac{6}{5}$

⑤  $\lim_{h \rightarrow 0} \frac{(4+h)^2 - 16}{h} = ?$   $\lim_{h \rightarrow 0} \frac{(4+h)^2 - 16}{h} = \lim_{h \rightarrow 0} \frac{(16 + 8h + h^2) - 16}{h} = \lim_{h \rightarrow 0} \frac{8h + h^2}{h} = \lim_{h \rightarrow 0} 8 + h = 8$

⑥  $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 - 1} = ?$   $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 - 1} = \lim_{x \rightarrow 1} \frac{(x-1)(x^2 + x + 1)}{(x-1)(x+1)} = \frac{3}{2}$

⑦  $\lim_{x \rightarrow 7} \frac{\sqrt{x+2} - 3}{x-7} = ?$   $\lim_{x \rightarrow 7} \frac{\sqrt{x+2} - 3}{x-7} \cdot \frac{(\sqrt{x+2} + 3)}{(\sqrt{x+2} + 3)} = \lim_{x \rightarrow 7} \frac{(x+2) - 9}{(x-7)(\sqrt{x+2} + 3)}$   
 $= \lim_{x \rightarrow 7} \frac{1}{(\sqrt{x+2} + 3)} = \frac{1}{6}$

⑧  $\lim_{x \rightarrow 2} \frac{x^4 - 16}{x-2} = ?$   $\lim_{x \rightarrow 2} \frac{(x^4 - 16)}{(x-2)} = \lim_{x \rightarrow 2} \frac{(x^2 - 4)(x^2 + 4)}{(x-2)} = \lim_{x \rightarrow 2} \frac{(x-2)(x+2)(x^2 + 4)}{(x-2)}$   
 $= \lim_{x \rightarrow 2} (x+2)(x^2 + 4) = 32$

⑨  $\lim_{x \rightarrow 1} \frac{\sqrt{x} - x^2}{1 - \sqrt{x}} = ?$   $\lim_{x \rightarrow 1} \frac{\sqrt{x} - x^2}{1 - \sqrt{x}} = \lim_{x \rightarrow 1} \frac{\sqrt{x}(1 - x^{3/2})}{1 - \sqrt{x}}$   
 $= \lim_{x \rightarrow 1} \frac{\sqrt{x}(1 - \sqrt{x})(1 + \sqrt{x} + x)}{(1 - \sqrt{x})} = \lim_{x \rightarrow 1} \sqrt{x}(1 + \sqrt{x} + x) = 3$

Devg  
Attintar

②  $\lim_{x \rightarrow 0} x^2 \cos(20\pi x) = ?$

Use squeeze theorem

$$-1 \leq \cos(20\pi x) \leq 1$$

$$-x^2 \leq x^2 \cos(20\pi x) \leq x^2$$

$$\lim_{x \rightarrow 0} (-x^2) = 0$$

$$\lim_{x \rightarrow 0} x^2 = 0$$

S.T say that

$$\lim_{x \rightarrow 0} x^2 \cos(20\pi x) = 0$$

③ If  $1 \leq f(x) \leq x^2 + 2x + 2$  for all  $x$ , find  $\lim_{x \rightarrow -1} f(x)$

$$\lim_{x \rightarrow -1} 1 = 1$$

$$\lim_{x \rightarrow -1} x^2 + 2x + 2 = -1$$

According to the squeeze

Theorem

$$\lim_{x \rightarrow -1} f(x) = 1$$

④ Prove that  $\lim_{x \rightarrow 0^+} \sqrt{x} e^{\sin(\pi/x)} = 0$

$$-1 \leq \sin(\pi/x) \leq 1$$

$$e^{-1} \leq e^{\sin(\pi/x)} \leq e^1$$

$$\sqrt{x} \cdot e^{-1} \leq \sqrt{x} e^{\sin(\pi/x)} \leq \sqrt{x} \cdot e^1$$

$$\frac{\sqrt{x}}{e} \leq \sqrt{x} e^{\sin(\pi/x)} \leq \sqrt{x} \cdot e$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{x}}{e} = \lim_{x \rightarrow 0} \sqrt{x} e = 0$$

Satz durch Theorem gives

$$\lim_{x \rightarrow 0} \sqrt{x} e^{\sin(\pi/x)} = 0$$

19. 20 21 22 29

5 Find the following limits if it exists

a)  $\lim_{x \rightarrow 1.5} \frac{2x^2 - 3x}{(2x-3)}$

b)  $\lim_{x \rightarrow 0} \left( \frac{1}{x} - \frac{1}{|x|} \right)$

a)  $\lim_{x \rightarrow 1.5^+} \frac{2x^2 - 3x}{(2x-3)} = \lim_{x \rightarrow 1.5^+} \frac{(2x-3)x}{(2x-3)} = 1.5$   
 $\lim_{x \rightarrow 1.5^-} \frac{2x^2 - 3x}{-(2x-3)} = \lim_{x \rightarrow 1.5^-} \frac{x(2x-3)}{-(2x-3)} = -1.5$   
 $1.5 \neq -1.5$   
 limit does not exist

b)  $\lim_{x \rightarrow 0^+} \left( \frac{1}{x} - \frac{1}{|x|} \right) = 0$

$\lim_{x \rightarrow 0^-} \left( \frac{1}{x} - \frac{1}{|-x|} \right) = \frac{2}{x} \rightarrow \infty$

Limit doesn't exist at  $x=0$

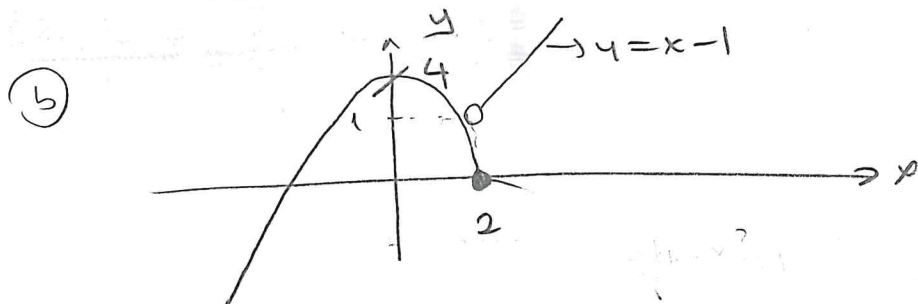
6 Let  $f(x) = \begin{cases} 4-x^2 & \text{if } x \leq 2 \\ x-1 & \text{if } x > 2 \end{cases}$

a) find  $\lim_{x \rightarrow 2^-} f(x)$  and  $\lim_{x \rightarrow 2^+} f(x)$

b) Does  $\lim_{x \rightarrow 2} f(x)$  exist?

c) Sketch the graph of  $f(x)$

a)  $\lim_{x \rightarrow 2^+} x-1 = 1$   $\lim_{x \rightarrow 2^-} 4-x^2 = 0$   $0 \neq 1$  limit doesn't exist.



7 If  $f(x) = \begin{cases} x^2 & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$  prove  $\lim_{x \rightarrow 0} f(x) = 0$

Observe that

$0 \leq f(x) \leq x^2$  for

all  $x \in \mathbb{R}$  so

$\lim_{x \rightarrow 0} 0 = 0$

$\lim_{x \rightarrow 0} x^2 = 0$

By squeeze theorem

$\lim_{x \rightarrow 0} f(x) = 0$

Deyan Altiton

8 Evaluate  $\lim_{x \rightarrow 2} \frac{\sqrt{6-x} - 2}{\sqrt{3-x} - 1} = ?$

$$\begin{aligned} & \lim_{x \rightarrow 2} \frac{\sqrt{6-x} - 2}{\sqrt{3-x} - 1} \cdot \frac{(\sqrt{6-x} + 2)}{(\sqrt{6-x} + 2)} \cdot \frac{(\sqrt{3-x} + 1)}{(\sqrt{3-x} + 1)} = \\ & = \lim_{x \rightarrow 2} \frac{((6-x) - 4)(\sqrt{3-x} + 1)}{((3-x) - 1)(\sqrt{6-x} + 2)} = \lim_{x \rightarrow 2} \frac{2-x}{2-x} \cdot \frac{\sqrt{3-x} + 1}{\sqrt{6-x} + 2} \\ & = \frac{2}{4} = \frac{1}{2} \end{aligned}$$

9 Is there a number  $a$  such that  $\lim_{x \rightarrow -2} \frac{3x^2 + ax + a + 3}{x^2 + x - 2}$  exists? If so, find the value of  $a$  and the value of the limit.  
 Since the denominator  $= 0$  at  $x = -2$ . To have limit, numerator must be also 0. Then,  $(x - (-2))$  will be factor of both numerator and denominator and they will be cancelled.

$$3x^2 + ax + a + 3 = 0 \quad \text{at } x = -2 \quad 3 \cdot 4 - 2a + a + 3 = 0$$

$$\underline{a = 15}$$

$$\text{Then, we get } \lim_{x \rightarrow -2} \frac{3x^2 + 15x + 18}{x^2 + x - 2} = \lim_{x \rightarrow -2} \frac{3[x^2 + 5x + 6]}{x^2 + x - 2}$$

$$\lim_{x \rightarrow -2} 3 \frac{(x+3)(x+2)}{(x+2)(x-1)} = \frac{3}{-3} = -1$$

10) Prove the following statements by using formal definition of limit?

a)  $\lim_{x \rightarrow 3} \frac{x}{5} = \frac{3}{5}$

e)  $\lim_{x \rightarrow 2} (x^2 - 1) = 3$

b)  $\lim_{x \rightarrow 6} \left( \frac{x}{4} + 3 \right) = 9/2$

f)  $\lim_{x \rightarrow 3} \frac{x^2 + x - 12}{x - 3} = 7$

c)  $\lim_{x \rightarrow -5} \left( 4 - \frac{3x}{5} \right) = 7$

g)  $\lim_{x \rightarrow 9^-} \sqrt[4]{\frac{x-3}{9-x}} = 0$

d)  $\lim_{x \rightarrow 2} (x^2 - 4x + 5) = 1$

h)  $\lim_{x \rightarrow 2} x^3 = 8$

a) Given  $\varepsilon > 0$ , we need to find  $\delta > 0$  such that if  $0 < |x - 3| < \delta$ , then  $\left| \frac{x}{5} - \frac{3}{5} \right| < \varepsilon$ .

$$\left| \frac{x}{5} - \frac{3}{5} \right| = \frac{1}{5} |x - 3| < \varepsilon \quad |x - 3| < 5\varepsilon$$

So choose  $\delta = 5\varepsilon$ . Then,  $|x - 3| < 5\varepsilon \Rightarrow \left| \frac{x}{5} - \frac{3}{5} \right| < \varepsilon$

b) Given  $\varepsilon > 0$   $\left| \frac{x}{4} + 3 - 9/2 \right| < \varepsilon$  search for  $\delta > 0$  for  $|x - 6| < \delta$

$$\left| \frac{x}{4} + \frac{6}{2} - \frac{9}{2} \right| = \left| \frac{x}{4} - \frac{3}{2} \right| = \left| \frac{x}{4} - \frac{6}{4} \right| < \varepsilon$$

$$|x - 6| < 4\varepsilon \quad \delta = 4\varepsilon$$

c) Given  $\varepsilon > 0$   $\left| 4 - \frac{3x}{5} - 7 \right| < \varepsilon$  obtain  $|x - (-5)| < \delta$

$$\left| -\frac{3x}{5} - 3 \right| = \left| -\frac{3x}{5} - \frac{15}{5} \right| = \left| -\frac{3}{5} (x + 5) \right| = \frac{3}{5} |x + 5| < \varepsilon$$

$$|x + 5| < \frac{5}{3} \varepsilon \quad \delta = \frac{5}{3} \varepsilon$$

Deyaj Attinta

d) Given  $\varepsilon > 0$ , we need to find  $\delta > 0$  such that  
if  $0 < |x-2| < \delta$ , then  $|(x^2-4x+5)-1| < \varepsilon$

$$|x^2-4x+4| < \varepsilon \quad |(x-2)^2| < \varepsilon \quad \text{so } |x-2| < \sqrt{\varepsilon}$$

$$\delta = \sqrt{\varepsilon}$$

e) Given  $\varepsilon > 0$ , we need to find  $\delta > 0$  such that if

$0 < |x-(-2)| < \delta$ , then  $|(x^2-1)-3| < \varepsilon$ . or upon  
simplifying we need  $|x^2-4| < \varepsilon$  whenever  $|x+2| < \delta$ .

$$|(x^2-4)| = |(x-2)(x+2)| < \varepsilon.$$

$$\text{Assume } \delta_1 = 1 \quad |x+2| < 1 \quad -1 < x+2 < 1$$

$$-1-4 < x-2 < 1-4 \quad -5 < x-2 < -3$$

Then, we get  $|x-2| < 5$ . This will give

$$|x-2||x+2| < 5 \cdot \delta < \varepsilon \quad \delta_2 = \varepsilon/5.$$

$$\delta = \min(\delta_1, \delta_2) = \varepsilon/5.$$

f) Given  $\varepsilon > 0$ , we need to find  $\delta > 0$  such that

$$\text{if } 0 < |x-3| < \delta, \text{ then } \left| \frac{x^2+x-12}{x-3} - 7 \right| < \varepsilon.$$

Notice that if  $x \neq 3$  we have  $\frac{x^2+x-12}{x-3} = x+4$ .

Then, when  $0 < |x-3| < \delta$ , we have  $|x+4-7| < \varepsilon$ .

$$|x-3| < \varepsilon. \quad \text{Take } \delta = \varepsilon.$$

Deja Attunten

9) Given  $\epsilon > 0$ , we need  $\delta > 0$  such that if  $\underbrace{9 - \delta < x < 9}_{x \rightarrow 9^-}$  then  $|\sqrt[4]{9-x} - 0| < \epsilon \Rightarrow 9-x < \epsilon^4$   $\therefore \epsilon > 0$ .

$$9 - \epsilon^4 < x < 9 \quad \text{Choose } \delta = \epsilon^4$$

Because  $9-x < \epsilon^4 > 0$   
 $9-x > 0 \quad x < 9$ .

10)  $\lim_{x \rightarrow 2} x^3 = 8$  Given  $\epsilon > 0$ , we need  $\delta > 0$  such that

if  $0 < |x-2| < \delta$ , then  $|x^3 - 8| < \epsilon$ .

$$|x^3 - 8| = |(x-2)(x^2 + 2x + 4)| < \epsilon$$

$$\text{Choose } \delta_1 = 1 \quad |x-2| < 1 \quad 1 < x < 3$$

$$\text{Since } x < 3 \quad x^2 + 2x + 4 < 9 + 2 \cdot 3 + 4 = 19$$

$$|x-2| |x^2 + 2x + 4| < 19 \cdot \delta < \epsilon \quad \delta_2 = \epsilon/19$$

$$\delta = \min(\delta_1, \delta_2) = \min(1, \epsilon/19)$$

11) Prove that  $\lim_{x \rightarrow 3} \frac{1}{(x+3)^4} = 0$

Given  $M > 0$ , we need  $\delta > 0$  such that  $0 < |x+3| < \delta$

$$\frac{1}{(x+3)^4} > M \quad (x+3)^4 < \frac{1}{M} \Leftrightarrow |x+3| < \frac{1}{\sqrt[4]{M}}$$

$$\text{Take } \delta = \frac{1}{\sqrt[4]{M}}$$

12)  $\lim_{x \rightarrow \infty} \ln x = \infty$

Given  $m < 0$  we need to find  $\delta > 0$  whenever  $\ln x < m$   $\underbrace{0 < x < \delta}_{x \rightarrow 0^+}$   $x = e^{\ln x}$   $e^{\ln x} < e^m$

Then, we take  $\delta = e^m$ .

Deyu Attanta

13 Suppose that  $\lim_{x \rightarrow a} f(x) = \infty$  and  $\lim_{x \rightarrow a} g(x) = c$  where  $c \in \mathbb{R}$

Prove that  $\lim_{x \rightarrow a} [f(x) + g(x)] = \infty$

Let  $M$  be given. Since  $\lim_{x \rightarrow a} f(x) = \infty$  there exists  $\delta_1$

$\delta_1 > 0$  such that  $0 < |x - a| < \delta_1 \implies f(x) > M + 1 - c$ .

we have an  $N$  instead of  $M$ .  
we write it w.r.t  $M, c$

Since  $\lim_{x \rightarrow a} g(x) = c$ , there exists  $\delta_2 > 0$  such that

$$|x - a| < \delta_2 \implies |g(x) - c| < 1 \implies -1 < g(x) - c < 1$$

$$c - 1 < g(x) < c + 1 \implies g(x) > c - 1$$

Then choosing  $\delta = \min(\delta_1, \delta_2)$ ,  $f(x) + g(x) > M + 1 - c + c - 1$

$f(x) + g(x) > M$  this means that  $\lim_{x \rightarrow a} [f(x) + g(x)] = \infty$

14 Calculate the following limits

a)  $\lim_{x \rightarrow \infty} \frac{3x^2 - x + 4}{2x^2 + 5x - 8} = ?$

$$\lim_{x \rightarrow \infty} \frac{x^2 (3 - \frac{1}{x} + \frac{4}{x^2})}{x^2 (2 + \frac{5}{x} - \frac{8}{x^2})} = \frac{3}{2}$$

b)  $\lim_{x \rightarrow \infty} \sqrt{\frac{12x^3 - 5x + 2}{1 + 4x^2 + 3x^3}} = \sqrt[3]{\lim_{x \rightarrow \infty} \frac{x^3 (12 - \frac{5}{x} + \frac{2}{x^2})}{x^3 (\frac{1}{x^3} + \frac{4}{x} + 3)}} = \sqrt[3]{12} = 2$

Peter Attwater

$$\textcircled{c} \lim_{x \rightarrow \infty} \frac{x^3 + 5x}{2x^3 - x^2 + 4} = ?$$

$$\lim_{x \rightarrow \infty} \frac{x^3 (1 + \frac{5}{x^2})}{x^3 (2 - \frac{1}{x} + \frac{4}{x^3})} = \frac{1}{2}$$

$$\textcircled{d} \lim_{x \rightarrow -\infty} \frac{t^2 + 2}{t^3 + t^2 - 1} = ? \quad \lim_{x \rightarrow -\infty} \frac{t^2 (\frac{1}{t} + 2(\frac{1}{t^3}))}{t^3 (1 + \frac{1}{t} - \frac{1}{t^3})} = \frac{2}{1} = 2$$

$$\textcircled{e} \lim_{x \rightarrow \infty} \frac{\sqrt{9x^6 - x}}{x^3 + 1} = ? \quad \lim_{x \rightarrow \infty} \frac{\frac{\sqrt{9x^6 - x}}{x^3}}{\frac{x^3 + 1}{x^3}} = \lim_{x \rightarrow \infty} \frac{\sqrt{9 \frac{x^6}{x^6} - \frac{x}{x^6}}}{1 + \frac{1}{x^3}}$$

$$= \lim_{x \rightarrow \infty} \frac{\sqrt{9 + \frac{1}{x^5}}}{1 + \frac{1}{x^3}} = 3$$

$$(x^3 = \sqrt{x^6})$$

$$\textcircled{f} \lim_{x \rightarrow \infty} (\sqrt{9x^2 + x} - 3x) = ?$$

$$\lim_{x \rightarrow \infty} \sqrt{9x^2 + x} - 3x \cdot \frac{(\sqrt{9x^2 + x} + 3x)}{(\sqrt{9x^2 + x} + 3x)} = \lim_{x \rightarrow \infty} \frac{9x^2 + x - 9x^2}{(\sqrt{9x^2 + x} + 3x)}$$

$$\lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2(9 + \frac{1}{x})} + 3x} = \lim_{x \rightarrow \infty} \frac{x}{|x| \sqrt{9 + \frac{1}{x}} + 3x} = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{9 + \frac{1}{x}} + 3} = \frac{1}{6}$$

$|x| = x$  for  $x > 0$   $x \rightarrow \infty$

$$\textcircled{g} \lim_{x \rightarrow -\infty} (x + \sqrt{x^2 + 2x}) = ?$$

Deyu Altun

$$\lim_{x \rightarrow -\infty} x + \sqrt{x^2 + 2x} \cdot \frac{(x - \sqrt{x^2 + 2x})}{(x - \sqrt{x^2 + 2x})} = \lim_{x \rightarrow -\infty} \frac{x^2 - (x^2 + 2x)}{x - \sqrt{x^2 + 2x}}$$

$$= \lim_{x \rightarrow -\infty} \frac{-2x}{x - \sqrt{x^2(1 + \frac{2}{x})}} = \lim_{x \rightarrow -\infty} \frac{-2x}{x - |x| \sqrt{1 + \frac{2}{x}}} = \lim_{x \rightarrow -\infty} \frac{-2}{1 - \sqrt{1 + \frac{2}{x}}} = \frac{1}{1}$$

(h)  $\lim_{x \rightarrow \infty} \cos x = ?$

This limit does not exist because as  $x$  increases  $\cos x$  doesn't approach any value, but oscillates between  $-1$  and  $1$ .

(I)  $\lim_{x \rightarrow -\infty} (x^4 + x^5) = ?$   $\lim_{x \rightarrow -\infty} x^5 \left(1 + \frac{1}{x}\right) = -\infty$

(J)  $\lim_{x \rightarrow \infty} \frac{x^3 - 2x + 3}{5 - 2x^2} = ?$   $\lim_{x \rightarrow \infty} \frac{x^3 (1 - \frac{2}{x^2} + \frac{3}{x^3})}{x^2 (\frac{5}{x^2} - \frac{2}{x})} = \infty$

(15) Prove  $\lim_{x \rightarrow \infty} \frac{1}{\sqrt{x}} = 0$

If  $\epsilon > 0$  is given, then  $|\frac{1}{\sqrt{x}} - 0| < \epsilon \Rightarrow \sqrt{x} > \frac{1}{\epsilon}$   
 $x > \frac{1}{\epsilon^2}$ . Let  $N = \frac{1}{\epsilon^2}$ . Then, when  $x > \frac{1}{\epsilon^2}$

$|\frac{1}{\sqrt{x}} - 0| < \epsilon$  so  $\lim_{x \rightarrow \infty} \frac{1}{\sqrt{x}} = 0$ .

(16) Prove that  $\lim_{x \rightarrow \infty} x^3 = \infty$

Given  $M > 0$ , we need  $N > 0$  such that  $x > N \Rightarrow x^3 > M$ .

Now  $x^3 > M \Leftrightarrow x > \sqrt[3]{M}$ . Then we take  $N = \sqrt[3]{M}$ .

(17) Prove that  $\lim_{x \rightarrow 0} f(x) = \lim_{t \rightarrow 0^+} f(1/t)$

Suppose that  $\lim_{x \rightarrow 0} f(x) = L$ . Then for every  $\epsilon > 0$ , there is

a corresponding  $N$  such that  $|f(x) - L| < \epsilon$  whenever  $x > N$ . If  $t = \frac{1}{x}$  Then  $x > N \Leftrightarrow 0 < \frac{1}{x} < \frac{1}{N}$

$0 < t < \frac{1}{N}$

Thus for every  $\epsilon > 0$  there is  $\frac{1}{N}$  such that

$|f(1/t) - L| < \epsilon$

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18 Find numbers at which points  $f$  is discontinuous

a 
$$f(x) = \begin{cases} 1+x^2 & \text{if } x \leq 0 \\ 2-x & \text{if } 0 < x \leq 2 \\ (x-2)^2 & \text{if } x > 2 \end{cases}$$

$x < 0$   $1+x^2 = f(x) \rightarrow$  a polynomial cont. for all  $(-\infty, 0)$

$x = 0$   $\lim_{x \rightarrow 0^-} 1+x^2 = 1$   $\lim_{x \rightarrow 0^+} 2-x = 2$   $1 \neq 2$

$\lim_{x \rightarrow 0} f(x)$  DOES NOT EXIST. NOT CONT at  $x = 0$

$0 < x < 2$   $f(x) = 2-x \rightarrow$  a polynomial cont. at this interval

$x = 2$   $\lim_{x \rightarrow 2^-} 2-x = 0$   $\lim_{x \rightarrow 2^+} (x-2)^2 = 0$   $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$

$\lim_{x \rightarrow 2} f(x) = 0$   $f(2) = 0$  CONT AT  $x = 2$

$x > 2$   $f(x) = (x-2)^2$  is a polynomial and cont at all points.

b 
$$f(x) = \begin{cases} x+2 & \text{if } x < 0 \\ e^x & \text{if } 0 \leq x \leq 1 \\ 2-x & \text{if } x > 1 \end{cases}$$

$x < 0$   $f(x) = x+2 \rightarrow$  a polynomial cont. everywhere.

$x = 0$   $\lim_{x \rightarrow 0^+} e^x = 1$   $\lim_{x \rightarrow 0^-} x+2 = 2$   $\lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$

$\lim_{x \rightarrow 0} f(x)$  DOES NOT EXIST. NOT CONT. at  $x = 0$

$0 < x \leq 1$  it is a polynomial cont everywhere

$x = 1$   $\lim_{x \rightarrow 1^-} e^x = e$   $\neq \lim_{x \rightarrow 1^+} 2-x = 1$

$\lim_{x \rightarrow 1} f(x)$  doesn't exist NOT CONT at  $x = 1$

$x > 1$   $f(x) = 2-x$  is cont everywhere



Find  $c$  that makes  $g$  continuous on  $(-\infty, \infty)$

$$g(x) = \begin{cases} x^2 - c^2 & x < 4 \\ cx + 20 & x \geq 4 \end{cases}$$

$$\lim_{x \rightarrow 4^-} x^2 - c^2 = 16 - c^2$$

$$\lim_{x \rightarrow 4^+} cx + 20 = 4c + 20$$

$$16 - c^2 = 4c + 20$$

$$c^2 + 4c + 4 = 0$$

$$(c+2)^2 = 0 \quad c = -2$$

If  $c = -2$ , then,  $\lim_{x \rightarrow 4} x^2 - 4 = 12 = f(4) = -2 \cdot 4 + 20 = 12$ .

for  $x < 4$   $x^2 - 4 = g(x)$  poly and cont. everywhere  
 $x \geq 4$   $-2x + 20 = g(x)$  poly and cont everywhere

20 If  $f(x) = x^3 - x^2 + x$ , show that there exist a number  $c$  such that  $f(c) = 10$

I. WAY  $x^3 - x^2 + x = 10 \Rightarrow$  It means that  $g(x) = x^3 - x^2 + x - 10$  has a root. Let's consider the time interval.

$$[1, 3] \quad g(1) = 1 - 1 + 1 - 10 = -9 < 0$$

$$g(3) = 27 - 9 + 3 - 10 = 11 > 0$$

This means that there exist a root in  $(1, 3)$   
 which means that.  $\rightarrow$  According to the Intermediate Value theorem

$$f(c) = 10 \text{ for } c \in (1, 3)$$

II. WAY

$$f(x) = x^3 - x^2 + x$$

$$f(1) = 1 - 1 + 1 = 1$$

$$f(3) = 27 - 9 + 3 = 21$$

Since  $f(1) < 10 < f(3)$  According to the Intermediate

Value Theorem  $f(c) = 10 \quad c \in (1, 3)$

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